Same Responses, Different Questions

AP Statistics · Unit 2 · Topic 2.1 · B: 2026-10-15 · E: 2026-10-28 · TEACHER KEY

CED TETHER

- **Skill 2.D** — Compare distributions or relative positions of points within a distribution.
- **EU UNC-1** — Graphical representations and statistics allow us to identify and represent key features of data.
- **LO UNC-1.P** — Compare numerical and graphical representations for two categorical variables. [Skill 2.D]
- **EK UNC-1.P.1** — Side-by-side bar graphs, segmented bar graphs, and mosaic plots are examples of bar graphs for one categorical variable, broken down by categories of another categorical variable.
- **EK UNC-1.P.2** — Graphical representations of two categorical variables can be used to compare distributions and/or determine if variables are associated.
- **EK UNC-1.P.3** — A two-way table, also called a contingency table, is used to summarize two categorical variables. The entries in the cells can be frequency counts or relative frequencies.
- **EK UNC-1.P.4** — A joint relative frequency is a cell frequency divided by the total for the entire table.

Essential question: Which display best answers a relationship question when the groups have different sizes?

DOK-3 skill: 4.B · **FRQ pattern:** `count-bars-vs-segmented-bars-which-answers-the-question`

DOK rationale: Part (c) requires matching two displays to different reader questions, using equal-height bars and conditional proportions to judge association, and explaining what information each encoding withholds.

Do Now (first take) → Explore (video + follow-along) → Exit (finish + turn in) **DOK:** 1 → 3 **Minutes:** 45

Teacher does

- Board slide up at the bell. Ask for a graph choice and one visible feature before students calculate.
- Begin the video follow-along at minute 5.
- Return to both graphs at minute 33. Circulate and keep the reader's exact question in view.
- Collect at the door. Score part (c) only, E/P/I.

Students do

- Choose a graph for the relationship question and cite one feature.
- Complete the video follow-along about two-way tables and bar-graph displays.
- Finish (a)–(c), revise the first take in the exit line, and turn in the sheet.

Questions to ask

- Why is the ninth-grade count bar twice as tall before preference is considered?
- What does making both grade bars end at 100 percent remove from the comparison?
- Which graph could answer how many students need Field Day supplies?
- What extra number would let you turn a segment percentage back into a count?

Adult role

Solo teacher. Circulate 5 pairs. Ask students to connect each graph to a specific reader question; accept both students' observations while pressing for which display answers association.

[WARN] WATCH FOR

- Reading equal total heights in Graph B as equal numbers of students.
- Comparing 72 with 18 without accounting for grade totals.
- Saying one graph is always better rather than matching counts or proportions to a purpose.

SCORING PART (c) — E / P / I

Expected elements

- **selects-segmented-for-association** (required) — Chooses Graph B for association and connects equal total heights to fair comparison of within-grade proportions
- **uses-conditional-contrast** (required) — Uses the 60 versus 30 percent Field Day contrast or the full conditional distributions as evidence
- **states-count-graph-purpose** (required) — Explains that Graph A preserves raw counts and unequal grade totals
- **gives-display-specific-limits** (required) — Gives a percentage question requiring calculation from Graph A and a count question impossible from Graph B without group sizes

E	Chooses Graph B with the equal-height proportion mechanism and numerical contrast, explains Graph A's count purpose, and gives one accurate limitation for each display.
P	Chooses Graph B and notes percentages but omits the equal-height mechanism, a numerical contrast, Graph A's purpose, or one display-specific limitation.
I	Chooses solely by appearance, treats the 100-percent bars as equal counts, or claims a segmented graph reveals the number surveyed.

Common mistakes

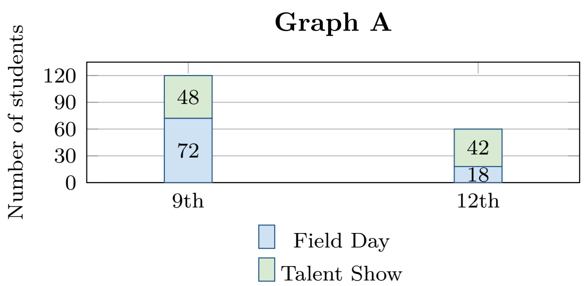
- Comparing 72 with 18 as if those counts were within-grade percentages
- Saying equal-height bars mean the two grade distributions are the same
- Claiming Graph A contains no information about percentages rather than saying division is required
- Claiming Graph B shows 120 ninth graders and 60 twelfth graders

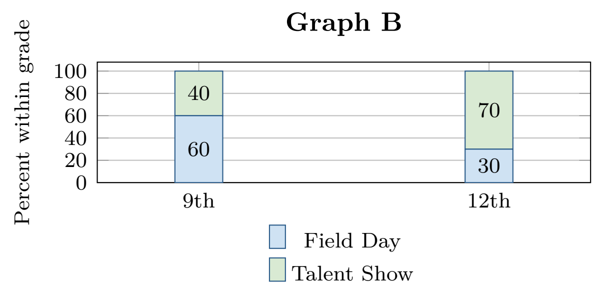
[STAR] TODAY'S PROBLEM — annotated

Suppose a student council asks 120 ninth graders and 60 twelfth graders whether they prefer Field Day or a Talent Show for the spring celebration. The same responses are shown in two graphs.

Andre: *Graph B hides how many people there are because both bars reach 100%.*

Leila: *Graph A hides the relationship because the grade totals are different.*





FIRST TAKE (5 min)

For the question *Is celebration preference related to grade level?*, which graph would you use first? Name one visible feature.
Key: Not graded. The goal is to surface whether students instinctively compare bar totals, segment counts, or within-grade shares.

Rules callout "MATCH THE DISPLAY TO THE QUESTION (keep on the page)" is printed on the student sheet.

DOK 1 **(a)** From Graph A, state each grade's total surveyed and the count in each preference category.

Key: Ninth grade: 120 total, with 72 preferring Field Day and 48 preferring Talent Show. Twelfth grade: 60 total, with 18 preferring Field Day and 42 preferring Talent Show.

DOK 2 **(b)** Compute the conditional distribution of preference within each grade. Describe the largest contrast between the grades.

Key: Ninth grade: Field Day $72/120 = 60\%$, Talent Show $48/120 = 40\%$. Twelfth grade: Field Day $18/60 = 30\%$, Talent Show $42/60 = 70\%$. Field Day preference is 30 percentage points higher in ninth grade; equivalently, Talent Show preference is 30 points higher in twelfth grade.

DOK 3 **(c)** For deciding whether preference is related to grade level, choose the more useful graph and explain how equal-height bars settle the comparison. State what Graph A is useful for. Then give one question that Graph A cannot answer *directly from bar lengths alone* and one question that Graph B cannot answer from the display.

Key: Graph B is more useful for the association question. Every grade bar has the same total height, so corresponding segment heights compare conditional proportions directly: 60%/40% for ninth grade versus 30%/70% for twelfth grade, clear evidence of association. Graph A is useful for reading raw counts and seeing that 120 ninth graders but only 60 twelfth graders were surveyed, so Andre identifies a real limitation of Graph B. Graph A cannot directly answer a percentage question such as What percent within each grade chose Field Day? without division. Graph B cannot answer a count question such as How many twelfth graders chose Talent Show? because it omits group sizes.

EXIT

One thing the video changed about my first take: ____________________